B. Patchouli's Magical Talisman

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: bitmasks, constructive algorithms, greedy, sortings | https://codeforces.com/problemset/problem/1688/B

She is skilled in all kinds of magics, and is keen on inventing new one.

—*Perfect Memento in Strict Sense*

Patchouli is making a magical talisman. She initially has n magical tokens. Their magical power can be represented with **positive** integers $a_1, a_2, \ldots, a_n$.

Patchouli may perform the following two operations on the tokens.

- **Fusion:** Patchouli chooses two tokens, removes them, and creates a new token with magical power equal to the sum of the two chosen tokens.
- **Reduction:** Patchouli chooses a token with an **even** value of magical power x, removes it and creates a new token with magical power equal to $x/2$.

Tokens are more effective when their magical powers are **odd** values. Please help Patchouli to find the minimum number of operations she needs to make magical powers of all tokens **odd** values.

Input

Each test contains multiple test cases.

The first line contains a single integer t ($1 \le t \le 10^3$) — the number of test cases. The description of the test cases follows.

For each test case, the first line contains one integer n ($1 \le n \le 2 \cdot 10^5$) — the initial number of tokens.

The second line contains n intergers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the initial magical power of the n tokens.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single integer — the minimum number of operations Patchouli needs to make all tokens have an **odd** value of magical power.

It can be shown that under such restrictions the required sequence of operations exists.

Example

Input
4 2 1 9 3 1 1 2 3 2 4 8 3 1049600 33792 1280

Output
0 1 3 10

Note

Test case 1:

a consists solely of odd numbers initially.

Test case 2:

Choose the tokens with magical power of 1 and 2 and perform Fusion. Now *a*=[1,3], both are odd numbers.

Test case 3:

Choose the tokens with magical power of 2 and 8 and perform Fusion. Now *a*=[4,10].

Choose the token with magical power of 10 and perform Reduction. Now *a*=[4,5].

Choose the tokens with magical power of 4 and 5 and perform Fusion. Now *a*=[9], and 9 is an odd number.

It can be shown that you can not make all the magical powers **odd** numbers in less than 3 moves, so the answer is 3.